

Project Interim Report
On
Contact Management Directory System
(Using Core Java & OOP Concepts)

RU-25-10631
Object Oriented Programming With Java

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Abstract

The Contact Directory Management System is a software-based application developed to efficiently store, manage, and retrieve contact information in a structured digital format. In traditional systems, contacts are often stored manually in notebooks or scattered files, which leads to difficulties in searching, updating, and maintaining data. This project aims to eliminate these issues by providing a centralized and user-friendly platform where users can perform operations such as adding, editing, deleting, and searching contact details with ease. The system ensures data consistency and minimizes redundancy by organizing information systematically. It also enhances productivity by reducing time spent on manual searches. The project demonstrates the use of programming concepts and data handling techniques to create a practical real-world application. Overall, this system offers a reliable and efficient solution for personal as well as professional contact management needs.

Introduction

In the modern digital era, the need for efficient data management systems has become increasingly important. Managing contact information manually is not only time-consuming but also prone to errors such as duplication, loss of data, and difficulty in retrieval. The Contact Directory Management System is designed to address these challenges by providing a digital solution for storing and managing contact details in an organized manner. This system allows users to store important information such as names, phone numbers, email addresses, and residential addresses in a structured database. Users can quickly search for specific contacts, update existing records, or delete outdated information. The system is designed with simplicity and usability in mind so that even users with minimal technical knowledge can operate it effectively. This project serves as a fundamental example of how software applications can simplify daily tasks and improve efficiency in handling personal or organizational data.

Scope

The scope of the Contact Directory Management System primarily focuses on providing essential functionalities required for managing contact information efficiently. The system allows users to add new contacts, view existing contacts, search for specific entries using keywords, update contact details, and delete unwanted records. It also provides sorting capabilities to arrange contacts alphabetically for better accessibility. The application is intended for individual users or small organizations that require a simple yet effective way to manage their contact data. Although the current version of the project is limited to basic operations, it has significant potential for future enhancements. Future developments may include features such as multi-user authentication, cloud-based storage, data backup and recovery, and integration with mobile devices. The system can also be expanded to support file import/export options and synchronization with external applications, making it more versatile and scalable.

System Requirements

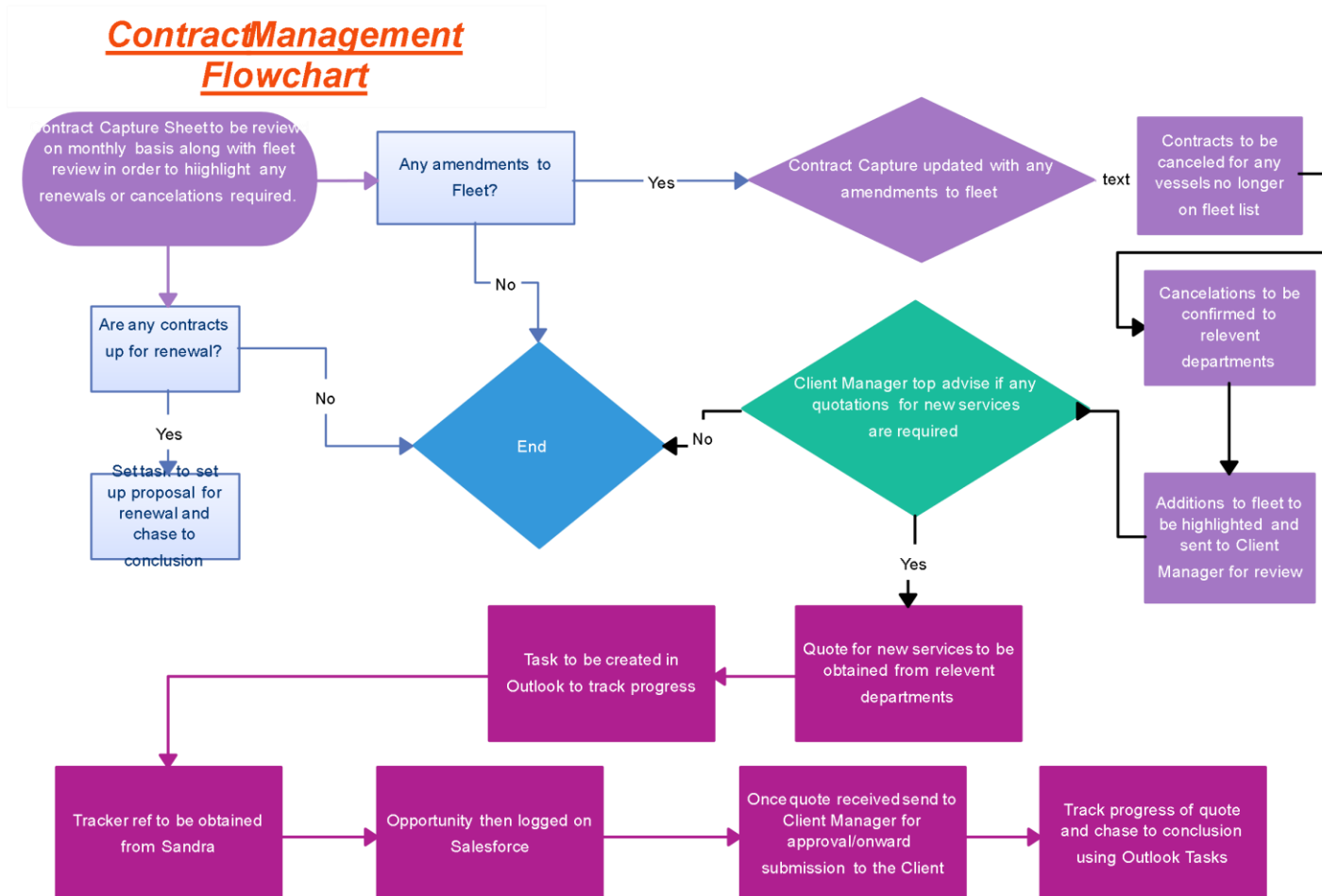
Hardware: Computer with 4GB RAM

Software: Java JDK, IDE (VS Code/Eclipse/Edit Plus)

Methodology

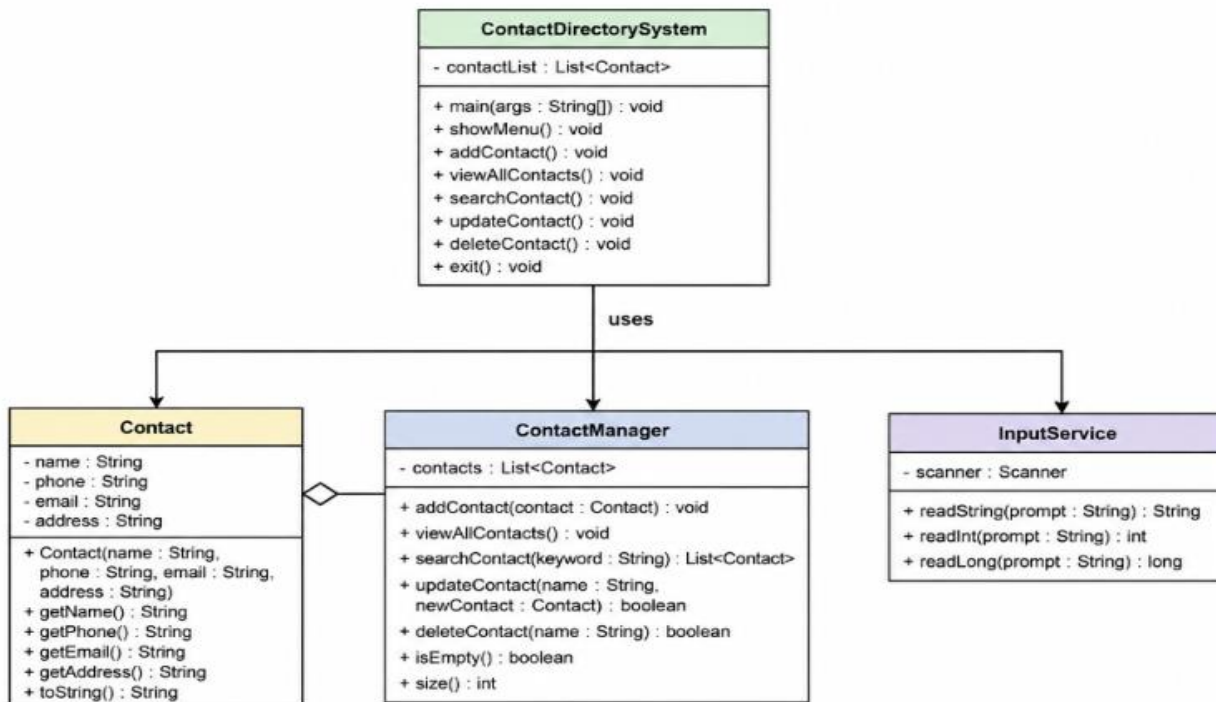
The development of the Contact Directory Management System follows a systematic and structured methodology to ensure efficiency and reliability. The first phase involves requirement analysis, where the needs and expectations of users are identified, such as storing, retrieving, and managing contact information. The second phase is system design, which includes designing the user interface and defining the structure of the data to be stored. This step ensures that the system is both user-friendly and logically organized. The third phase is implementation, where the actual coding of the application is done using Core Java by implementing Object-Oriented Programming concepts such as classes, objects, encapsulation, and method. The fourth phase is testing, where the system is evaluated for errors, bugs, and performance issues to ensure smooth functioning. Finally, the deployment phase involves running the system in a real environment where users can interact with it. This structured approach ensures the development of a reliable and efficient application.

Flowchart



Class Diagram

CLASS DIAGRAM – CONTACT DIRECTORY MANAGEMENT SYSTEM



Gantt Chart

1. Planning & Structure

A Gantt chart helps you break the project into tasks (design, coding, testing, etc.) and set timelines.

Even if it's already built once, rebuilding or modifying still needs planning.

2. Project Schedule Table

Task	Week 1	Week 2	Week 3	Week 4
Planning	✓			
Design	✓	✓		
Coding		✓	✓	
Testing			✓	
Documentation				✓

implementation

The project consists of multiple Java classes. The logic of each class is explained below (source code is not included as per report guidelines):

1. Contact Class

This class represents a single contact in the directory. It stores details such as contact name (String), phone number (String), email address (String), and address (String). The data members are kept private to ensure **encapsulation**, and getter and setter methods are used to access and modify the information. This class acts as the basic data model for the entire application.

2. ContactManager Class

This class manages multiple **Contact** objects. It stores a collection (ArrayList or array) of contacts and provides methods such as **addContact()**, **viewContacts()**, **searchContact()**, **updateContact()**, and **deleteContact()**. It acts as the core component of the system by handling all contact management operations efficiently.

3. Service Interface

This interface defines common operations related to contact management. It may include methods such as **searchContact()** or **displayContacts()**, which can be implemented by the **ContactManager** class. The use of an interface provides flexibility and demonstrates the implementation of **abstraction** in Object-Oriented Programming.

4. ContactDirectorySystem (Main Class)

This is the main class of the program where execution begins. It creates objects of the **Contact** and **ContactManager** classes and displays a menu-driven interface for user interaction. Based on the user's choice, it invokes methods such as **addContact()**, **viewContacts()**, **searchContact()**, **updateContact()**, and **deleteContact()** to perform the required operations. The class controls the overall workflow of the application and integrates all project modules.

Sample input/output

```
Enter Choice: 1
Enter Name: supriya singh
Enter Phone: 8886743963
Enter Email: supriya123@gmail.com
Contact Added Successfully!
```

```
--- Contact Directory ---
1. Add Contact
2. View Contacts
3. Search Contact
4. Update Contact
5. Delete Contact
6. Exit
Enter Choice: █
```

PROBLEMS 12 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Enter Name: supriya singh
Enter Phone: 8886743963
Enter Email: supriya123@gmail.com
Contact Added Successfully!
```

```
--- Contact Directory ---
1. Add Contact
2. View Contacts
3. Search Contact
4. Update Contact
5. Delete Contact
6. Exit
```

Conclusion

The Contact Directory Management System successfully fulfills its objective of providing a simple and efficient way to manage contact information digitally. It eliminates the limitations of manual record-keeping by offering quick access, easy modification, and organized storage of data. The system demonstrates the practical application of programming concepts such as data structures, functions, and user interaction. Through this project, it becomes evident how technology can simplify everyday tasks and improve productivity. Although the current system includes only basic functionalities, it serves as a strong foundation for future improvements and advanced features. Enhancements such as cloud integration, security mechanisms, and mobile compatibility can further increase its usability and effectiveness. Overall, this project not only provides a useful tool for managing contacts but also contributes to the learning and development of programming and problem-solving skills.

The project successfully demonstrates Java OOP concepts and provides a functional system.

References

Book

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